


Aviation Risk Analysis: Identifying Safer Aircraft Models for Purchase Recommendations

Phase 1 Project Presentation
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Overview

- Our company is expanding into aviation with plans for both commercial and private aircraft operations.
- Management requires clear, data-driven guidance on which models have the best safety records.
- This project analyzes accident data from 1919–2023 to identify safer options and provide recommendations.



Business Understanding

- *Business Problem:* Limited knowledge of aircraft safety risks creates uncertainty in purchasing decisions.
- *Goal:* Use historical accident data to identify patterns and guide aircraft investments.
- *Business Questions:*
 - Which aircraft models/types appear safer?
 - What are the accident trends over time?
 - Which operators and countries account for the most accidents?



Data Understanding

- *Source:* Kaggle – Aviation Accident Database (1919–2023).

<https://www.kaggle.com/drealbash/aviation-accident-from-1919-2023>

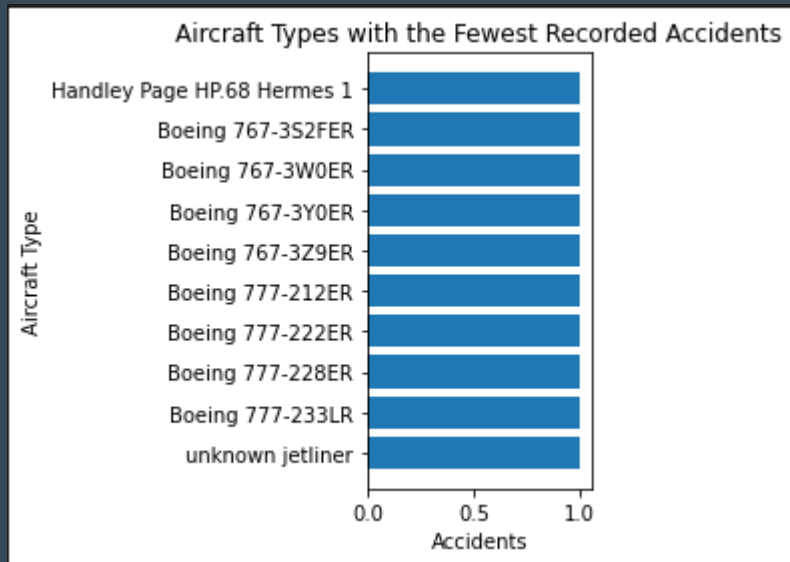
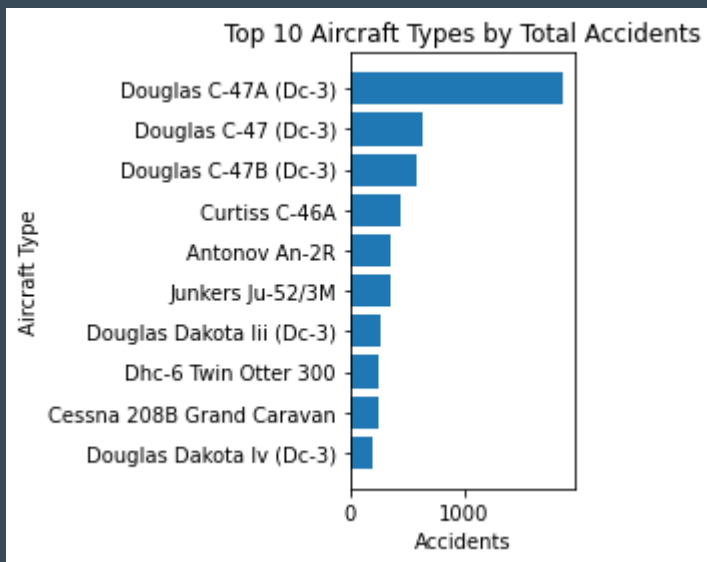
- *Size:* 22,887 rows × 8 columns after data cleaning.
- *Methods:* Removed duplicates, filled or dropped missing values, standardized fields, and aggregated accident counts.
- *Tools:* Python with pandas, matplotlib, and seaborn for analysis and visualization.





Data Analysis

Aircraft Models/Types: Accident Comparison and Insights



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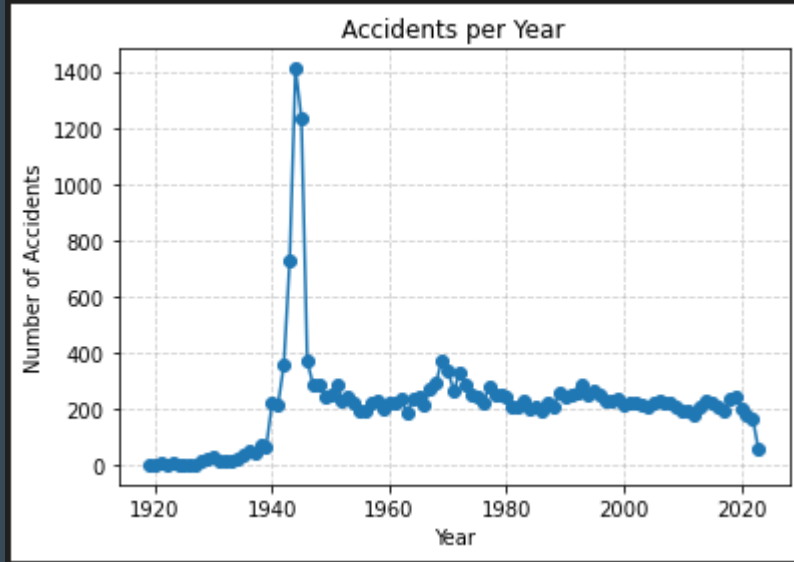
- Douglas C-47 family dominates accidents → avoid for purchase.
- Boeing 767 and Boeing 777 show very few accidents → safer alternatives.
- Clear gap between older high-risk models and newer safer designs.





Aviation Accident Trends Over Time

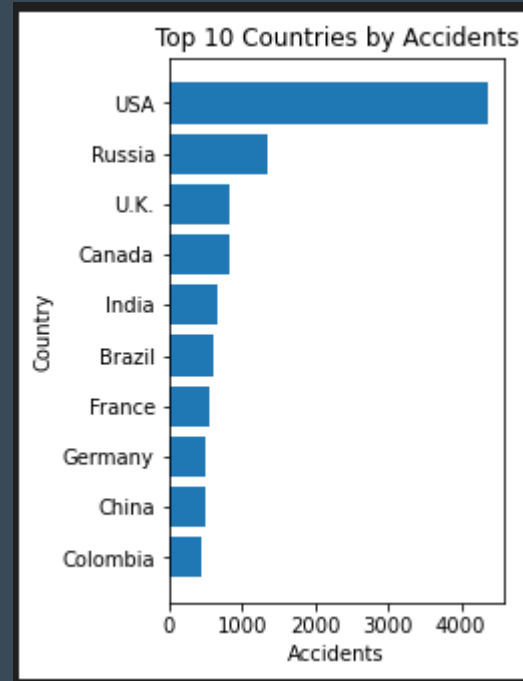
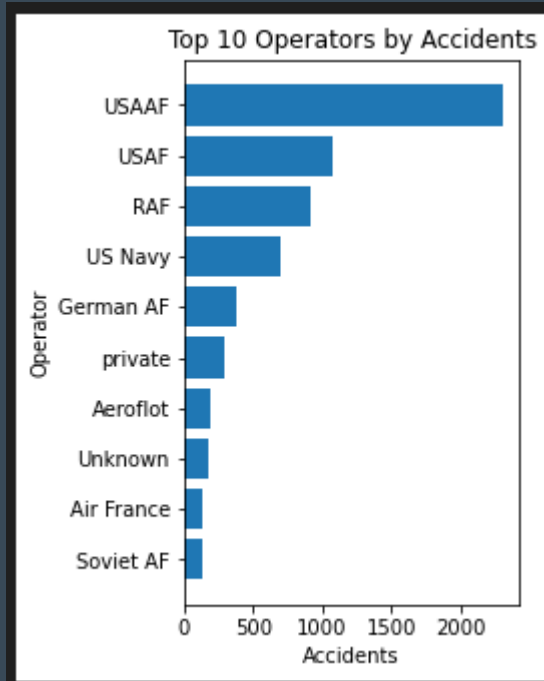
Accidents per year, 1919–2023



- Accidents peaked in 1940s, declined in 1950s.
- Stable counts in later decades.
- Drop in 2020s reflects improved safety standards.



Operators and Countries Accident Counts



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- Military operators (USAAF, USAF, RAF, US Navy) dominate accidents.
- Countries with highest accidents: USA, Russia, U.K.
- Safer to target commercial/private operators (e.g. Air France and Aeroflot) and lower-risk regions.





Recommendations





Business Recommendations for Safer Investments

01

- Avoid high-accident models such as the *Douglas C-47 family*, and prioritize modern aircraft with stronger safety records like the *Boeing 767* and *Boeing 777*.

02

- Focus investments on newer fleets that reflect decades of safety improvements, rather than older models that appear frequently in accident data.

03

- Work with established commercial/private operators (e.g. Air France and Aeroflot) and explore regions with lower accident rates such as China, France, and Germany.



Next Steps

- Run a financial and operational feasibility study to align safety choices with business goals.
- Engage with commercial/private operators in lower-risk markets.
- Assess regulations in safer regions like China, France, and Germany.



Thank You!

Thank you for your attention.

I welcome your questions and feedback.

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